COM3031 Probabilistic Machine Learning: Week 3

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Dr Zeyu Fu, Lecturer in Computer Vision and Machine Learning

z.fu@exeter.ac.uk

In this week

Lectures

- Why Approximation Inference
- Laplace Approximation
- Model Comparison and BIC

Workshop:

- Maximum A Posteriori

Part I: Why Approximation Inference

What do we learn from Bayesian Inference

Beyond a single "best" model

- In many ML methods, we estimate one best parameter value $\hat{\theta}$
- Bayesian inference instead gives us a distribution: $p(\theta \mid \mathcal{D})$, $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{n}$

What the posterior gives us

- Parameter uncertainty
  - How confident are we about different values of θ?
- Predictive uncertainty
  - How uncertain are predictions for new data?
- Robustness with small datasets
  - Priors help prevent extreme or unstable estimates

Why Approximate Inference?

Our goal is to infer the **posterior distribution over model parameters**: $p(\theta \mid \mathcal{D})$

Bayes' rule:

$$p(\theta \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \theta)p(\theta)}{p(\mathcal{D})}$$

The denominator is the **model evidence**:

$$p(\mathcal{D}) = \int p(\mathcal{D} \mid \theta)\, p(\theta)\, d\theta$$

- This integral is high-dimensional, often no closed-form solution and computationally expensive or impossible.

Exact Bayesian inference requires computing integrals that are usually intractable.

To use Bayesian methods in practice, we often need **approximate inference**.

Bayesian Linear Regression – A Rare Exact Case

Bayesian Linear Regression:

Model: $y = Xw + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2 I)$

Likelihood (Gaussian): $p(\,y \mid X, w, \sigma^2\,) = \mathcal{N}(\,y \mid Xw, \sigma^2 I\,)$

Prior on weights (Gaussian): $p(w) = \mathcal{N}(\,w \mid w_0, \Sigma_0\,)$

Posterior: Closed-Form Solution

- Because Gaussian prior + Gaussian likelihood:

$$p(\,w \mid y, X\,) = \mathcal{N}(\,w \mid w_n, \Sigma_n\,)$$

where

$$\Sigma_n^{-1} = \Sigma_0^{-1} + \frac{1}{\sigma^2} X^\top X, \qquad w_n = \Sigma_n \left(\Sigma_0^{-1} w_0 + \frac{1}{\sigma^2} X^\top y \right)$$

Special Case:

- Posterior can be computed exactly
- No approximations required
- But this relies crucially on *Gaussian–Gaussian conjugacy*.

When Exact Inference Breaks

Bayesian Logistic Regression

- Binary outputs: $y_i \in \{0,1\}$
- Likelihood: $p(\, y_i \mid x_i, w \,) = \text{Bernoulli}\big(\sigma(w^\top x_i)\big)$, where $\sigma(z) = \frac{1}{1+e^{-z}}$
- Prior on weights (same as before): $p(w) = \mathcal{N}(\, w \mid w_0, \Sigma_0 \,)$

The Posterior we want

$$p(\, w \mid D \,) \propto p(\, y \mid X, w \,)p(w)$$

Taking logs,

$$\log p(\, w \mid D \,) = \underbrace{\sum_{i=1}^{n} \left[y_i \log \sigma(w^\top x_i) + (1 - y_i) \log\big(1 - \sigma(w^\top x_i)\big) \right]}_{\text{log-likelihood}} + \underbrace{\log p(w)}_{\text{prior}}$$

What goes wrong?

- The sigmoid function makes the likelihood **non-Gaussian**
- The posterior: is **not Gaussian,** has **no closed-form normalising constant** And cannot be integrated analytically

The Core Idea of Approximate Inference

Instead of computing the exact posterior: $p(\theta \mid \mathcal{D})$, we approximate it with a simpler distribution: $p(\theta \mid \mathcal{D}) \approx q(\theta)$

Main families of approximate inference

- **Local approximations**
  - Approximate the posterior *near its peak*
  - Example: **Laplace approximation**
- **Optimisation-based methods**
  - Find the "best" approximating distribution
  - Example: **Variational inference**
- **Sampling-based methods**
  - Draw samples from the posterior
  - Example: **MCMC**

Part II: Laplace Approximation

Laplace Approximation: 1-D

To build intuition, start with a single continuous parameter: $\theta \in \mathbb{R}$

From Bayes' rule, the **posterior** has the form:

$$p(\theta \mid \mathcal{D}) = \frac{1}{Z}\tilde{p}(\theta), \qquad \tilde{p}(\theta) = p(\mathcal{D} \mid \theta)p(\theta)$$

- The normalisation constant is: $Z = \int \tilde{p}(\theta)d\theta$

Key observations:

- Z is usually **unknown / hard** to compute
- $\tilde{p}(\theta)$is an **unnormalised posterior**
- $\tilde{p}(\theta)$and $p(\theta \mid \mathcal{D})$ have the **same shape**
- They share the same mode (**MAP**): $\hat{\theta} = \arg\max_{\theta} p(\theta \mid \mathcal{D}) = \mathrm{argmax}_{\theta}\tilde{p}(\theta)$

Laplace approximation focuses on local shape near the mode $\hat{\theta}$, so we can ignore the constant Z while building the approximation.

Laplace Approximation: 1-D

Core idea: approximate a complicated posterior near its mode/peak using a Gaussian:

$$p(\theta \mid \mathcal{D}) \approx q(\theta) = \mathcal{N}\left(\theta \mid \hat{\theta}, \sigma^2\right)$$

- **Mean:** $\hat{\theta}$= **MAP estimate** (posterior mode)
- **Variance:** comes from **local curvature** at the mode
- **Interpretation:** sharper peak ⇒larger curvature ⇒smaller σ^2.

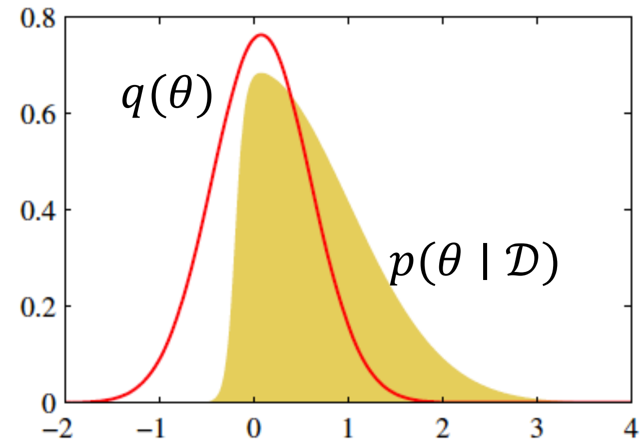

Image credits to the book of PRML by Bishop

Laplace Approximation: 1-D

Key move: Define the $\boldsymbol{log}$ **unnormalised posterior**

$$\tilde{p}(\theta) = p(\mathcal{D} \mid \theta)p(\theta) \qquad\qquad g(\theta) = \log \tilde{p}(\theta)$$

Second-order Taylor expansion around the mode $\hat{\theta}$:

$$g(\theta) \approx g(\hat{\theta}) + g'(\hat{\theta})(\theta - \hat{\theta}) + \frac{1}{2}g''(\hat{\theta})(\theta - \hat{\theta})^2$$

At the **mode/maximum**: $g'(\hat{\theta}) = 0$. So the linear term disappears, leaving:

$$g(\theta) \approx g(\hat{\theta}) + \frac{1}{2}g''(\hat{\theta})(\theta - \hat{\theta})^2$$

Since $\hat{\theta}$ is a maximum:

$$g''(\hat{\theta}) < 0$$

Define the **positive curvature**:

$$A = -g''(\hat{\theta})$$

Then:

$$g(\theta) \approx g(\hat{\theta}) - \frac{1}{2}A(\theta - \hat{\theta})^2$$

Laplace Approximation: 1-D

We obtained a **quadratic approximation** to the log unnormalised posterior:

$$g(\theta) = \log p(\mathcal{D} \mid \theta) + \log p(\theta) \approx g(\hat{\theta}) - \frac{1}{2}A(\theta - \hat{\theta})^2$$

where $A = -\frac{d^2 g(\theta)}{d\theta^2}\Big|_{\theta=\hat{\theta}} > 0$

Exponentiating both sides:

$$p(\theta \mid \mathcal{D}) \propto \exp\left(-\frac{1}{2}A(\theta - \hat{\theta})^2\right)$$

This has exactly the form of a **Gaussian density**. Therefore, the posterior is approximated by:

$$p(\theta \mid \mathcal{D}) \approx q(\theta) = \mathcal{N}\left(\theta \mid \hat{\theta}, A^{-1}\right)$$

- **Mean**: $\hat{\theta}$ (MAP estimate)
- **Variance**: A^{-1}(inverse curvature)

Laplace approximation turns local curvature of the log-posterior into a Gaussian uncertainty estimate.

Laplace Approximation: 1D

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Worked Exercise:

Consider the (unnormalised) density:

$$p(\theta) \propto \theta^{\alpha-1} \exp(-\beta\theta), \qquad \theta > 0$$

- This is proportional to a **Gamma distribution**
- Parameters: $\alpha > 0, \beta > 0$

- (On board) We want to **approximate** $p(\theta)$ using a **Gaussian distribution** via the **Laplace approximation**. That is, find:

$$p(\theta) \approx \mathcal{N}\left(\theta \mid \hat{\theta}, \sigma^2 \right)$$

Extension to Multiple Dimensions

From 1D to D-dimensions

Now consider a **parameter vector**: $\boldsymbol{\theta} \in R^D$, with posterior:

$$p(\boldsymbol{\theta} \mid \mathcal{D}) \;\propto\; p(\mathcal{D} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta})$$

As before, we work with the **log unnormalised posterior**:

$$g(\boldsymbol{\theta}) \;=\; \log p(\mathcal{D} \mid \boldsymbol{\theta}) \;+\; \log p(\boldsymbol{\theta})$$

Step 1: Find the mode (MAP): $\widehat{\boldsymbol{\theta}} = \arg\max_{\boldsymbol{\theta}} g(\boldsymbol{\theta})$

- Typically found using optimisation (e.g. gradient ascent / Newton's method)

Step 2: Local quadratic approximation:

- Use a **second-order Taylor expansion** of $g(\boldsymbol{\theta})$around $\widehat{\boldsymbol{\theta}}$:

$$g(\boldsymbol{\theta}) \approx \; g(\widehat{\boldsymbol{\theta}}) - \frac{1}{2}(\boldsymbol{\theta} - \widehat{\boldsymbol{\theta}})^{\top} \mathbf{H}\,(\boldsymbol{\theta} - \widehat{\boldsymbol{\theta}})$$

- where: $\mathbf{H} \;=\; -\nabla^2 g(\boldsymbol{\theta})|_{\boldsymbol{\theta}=\widehat{\boldsymbol{\theta}}} \quad \text{(negative Hessian)}$

Extension to Multiple Dimensions

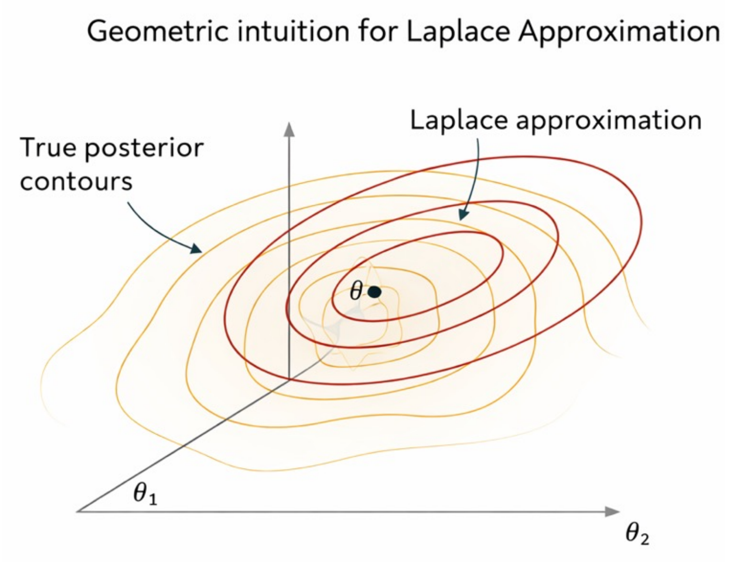
Step 3: Gaussian approximation

Exponentiating gives:

$$p(\boldsymbol{\theta} \mid \mathcal{D}) \approx \mathcal{N}(\boldsymbol{\theta} \mid \hat{\boldsymbol{\theta}}, \mathbf{H}^{-1})$$

Interpretation

- **Mean**: $\hat{\boldsymbol{\theta}}$ (MAP)
- **Covariance**: $\mathbf{H}^{-1}$
- Strong curvature → small uncertainty
- Flat curvature → large uncertainty



In multiple dimensions, Laplace approximation turns local curvature of the log-posterior into a full Gaussian covariance matrix.

Example: Logistic Regression

Recall logistic regression posterior:

$$p(w \mid \mathcal{D}) \propto \prod_i p(y_i \mid x_i, w), p(w)$$

- Posterior is **not Gaussian**
- No closed-form solution

Laplace approach:

- Find MAP $\hat{w}$ (same as regularised logistic regression)
- Compute Hessian of log posterior at $\hat{w}$
- Approximate posterior with:

$$p(w \mid \mathcal{D}) \approx \mathcal{N}(w \mid \hat{w}, \mathbf{H}^{-1})$$

This gives:

- Parameter uncertainty
- Predictive uncertainty (approximate)

Summary of Laplace Approx.

Works well when:

- Posterior is unimodal
- Posterior is approximately symmetric
- Data size is moderate to large
- Laplace is the simplest bridge from optimisation to Bayesian uncertainty.

Works poorly when:

- Posterior is multimodal
- Posterior is highly skewed
- Strong non-linearities (e.g. deep neural networks)
- Laplace is **local** — it only sees the peak, and may be limited to capturing important global perspectives.

Part III: Bayesian Model Comparison

Why Do We Compare Models?

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The problem: goodness of fit is not enough

- In many models, increasing complexity always improves fit to training data
- More parameters → more flexibility → lower training error
- But better fit does not mean better generalisation

Each panel fits the *same data* with increasing model complexity M

- M = 0,1: **Underfitting** — model too simple, misses structure
- M = 2,3: **Reasonable fit** — captures the main trend
- M ≥5: **Overfitting** — fits noise, unstable predictions

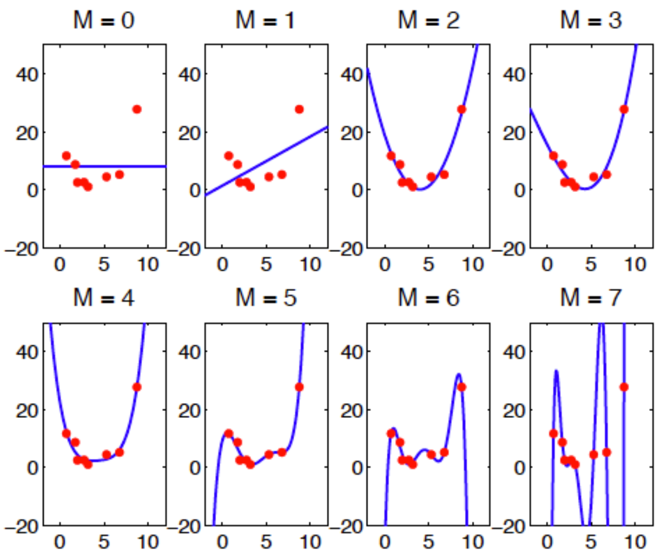


Image credit to: Unsupervised Learning Bayesian Model Comparison Zoubin Ghahramani, UCL,2005

Bayesian Model Comparison

Why likelihood alone fails

- Maximum likelihood prefers models that explain the data as closely as possible. This often means:
  - Very large parameters
  - Highly flexible functions
  - Poor predictions on new data
- Likelihood rewards fit, but ignores complexity

What we really want

- A good model should:
  - Fit the data **well**
  - Be **as simple as possible**
  - Generalise to **new, unseen data**
- This is the **bias–variance trade-off**, seen visually in the figure.

Bayesian perspective

- Bayesian model comparison balances:
  - **Data fit** (likelihood)
  - **Model complexity** (number of parameters, flexibility)
- This balance appears naturally through the **model evidence** $p(\mathcal{D} \mid \mathcal{M})$

Bayesian Model Comparison

Given several candidate models, which one should we prefer?

$$\mathcal{M}_1, \mathcal{M}_2, \dots, \mathcal{M}_K$$

- **Bayesian answer: compare model probabilities**
- Using Bayes' rule:

$$p(\mathcal{M} \mid \mathcal{D}) \propto p(\mathcal{D} \mid \mathcal{M}) p(\mathcal{M})$$

  - $p(\mathcal{M})$: prior belief in the model
  - $p(\mathcal{D} \mid \mathcal{M})$: **model evidence** (marginal likelihood)

If we assume all models are equally likely *a priori*:

$$p(\mathcal{M} \mid \mathcal{D}) \propto p(\mathcal{D} \mid \mathcal{M})$$

So **model comparison reduces to comparing evidences**.

Bayesian Model Comparison

What is model evidence?

$$p(\mathcal{D} \mid \mathcal{M}) = \int p(\mathcal{D} \mid \theta, \mathcal{M}) p(\theta \mid \mathcal{M}) d\theta$$

Why is it important?

- Model evidence decides which model is preferred
- It averages the likelihood over all parameter values
- Not just “best fit”, but overall explanatory power

Evidence balances two forces:

- Data fit: Can the model explain the data well?
- Model complexity**:** Does the model spread probability mass too widely?

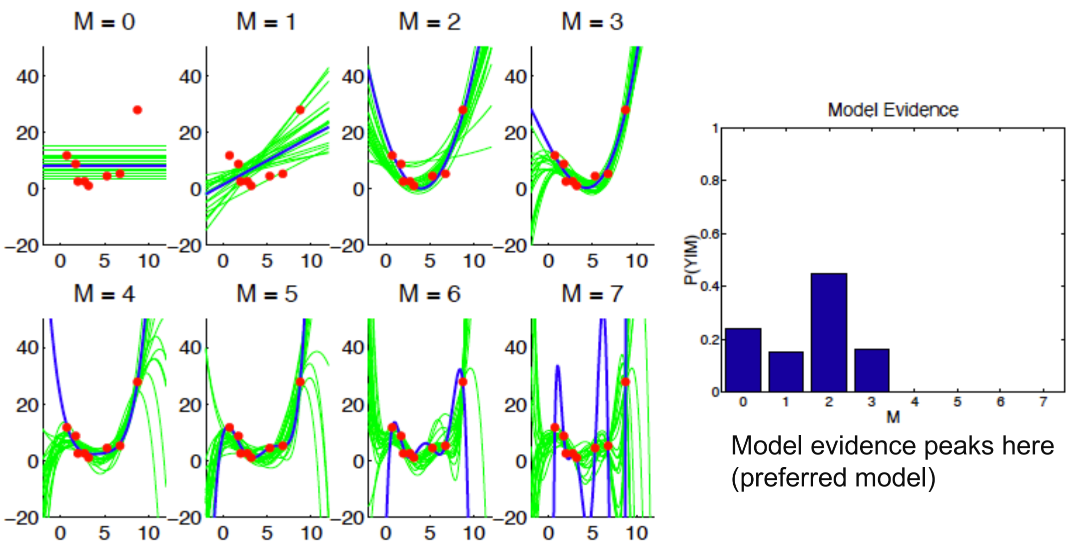
$$p(\mathcal{D} \mid \mathcal{M}) \approx p\left(\mathcal{D} \mid \hat{\theta}, \mathcal{M}\right) \times \text{Occam factor}$$

Occam's Razor via Model Evidence

Occam's razor (automatic, not manual)

- Simple models:
  - Concentrate probability mass
  - Explain data well over a **large region** of parameters
- Complex models:
  - Spread probability mass thinly
  - Fit data well only for finely tuned parameters

Bayesian evidence automatically penalises unnecessary complexity



Model evidence peaks here (preferred model)

Image credit to: Unsupervised Learning Bayesian Model Comparison Zoubin Ghahramani, UCL,2005

Model Comparison and BIC

Recall:

$$p(\mathcal{D} \mid \mathcal{M}) = \int p(\mathcal{D} \mid \theta, \mathcal{M}) p(\theta \mid \mathcal{M}) d\theta$$

Problems: High-dimensional θ, no closed form (logistic regression, neural nets) and computationally expensive

Laplace Approximation to the Evidence:

- Using Laplace around the MAP $\hat{\theta}$:

$$p(\mathcal{D} \mid \mathcal{M}) \approx p\left(\mathcal{D} \mid \hat{\theta}\right) p\left(\hat{\theta}\right) (2\pi)^{k/2} |\mathbf{H}|^{-1/2}$$

  - where k is number of parameters and $\mathbf{H}$ is Hessian of the negative log posterior

Model Comparison and BIC

Taking logs:

$$\log p(\mathcal{D} \mid \mathcal{M}) \approx \log p\big(\mathcal{D} \mid \hat{\theta}\big) - \frac{k}{2} \log n + \text{constants}$$

This leads directly to **Bayesian Information Criterion (BIC):**

$$\text{BIC} = \text{-}2 \log p\big(\mathcal{D} \mid \hat{\theta}\big) + k \log n$$

- $\hat{\theta}$: MLE or MAP
- k: number of parameters
- n: number of data points

Interpretation:

- First term: **data fit**
- Second term: **complexity penalty**
- Lower BIC = better model

Summary of BIC

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Strengths:

- Very cheap to compute
- Strong theoretical grounding
- Approximates Bayesian model comparison
- Encourages simpler models

Limitations:

- Asymptotic (assumes large n)
- Assumes unimodal posterior
- Based on Laplace (Gaussian) approximation
- Not suitable for highly flexible models